A.1 Foundations, Gradient descent, and backpropagation


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Welcome To Modern AI Summer Workshop!

Time (PM)	Day 1 Monday, July 6th	Day 2 Tuesday, July 7th	Day 3 Wednesday, July 8th	Day 4 Thursday, July 9th	Day 5 Friday, July 10th
12:00-1:00					**Keynote Speaker** Jason Tyler Rolfe Co-founder, CTO, Variational AI
1:00-1:30					Coffee Break
1:30-2:15	**Lecture A.1** Foundations, Gradient descent, and backpropagation	**Lecture B.4** Graph-structured data: GNNs	**Lecture D.2** Representation Learning: training and evaluation	**Lecture E.3** Generative modeling: diffusion models I	**Lecture F.1** Introduction to Multimodal Representation & Alignment
2:15-2:30	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break
2:30-3:15	**Lecture B.1** Tabular data: MLPs	**Lecture C.1** Attention and Transformer architectures	**Guest Speaker** Steven ten Holder Co-Founder & CEO Zeroshot Biolabs	**Guest Speaker** Carl de Boer Associate Professor UBC	**Lecture F.2** Multimodal Fusion
3:15-3:30	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break
3:30-4:00	**Tutorial 1:** Basics of Pytorch	**Tutorial 3:** CNNs	**Tutorial 5:** Contrastive learning	**Tutorial 7:** VAEs	**Tutorial 9:** Multimodal
4:00-4:15	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break
4:15-5:00	**Lecture B.2** Grid-structured data: CNNs	**Lecture C.2** Transformer: Application and training	**Lecture E.1** Generative modeling: concepts	**Lecture E.4** Generative modeling: diffusion models II	**Lecture F.3** Cross modal Learning
5:00-5:15	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Closing remarks
5:15-6:00	**Lecture B.3** Sequential data: sequence models	**Lecture D.1** Representation Learning: concepts	**Lecture E.2** Generative modeling: VAE	**Lecture E.5** Generative modeling: Structured (conditional) predictions	
6:00-6:30	**Tutorial 2:** Basics of Pytorch	**Tutorial 4:** Attention	**Tutorial 6:** Evaluating Representations	**Tutorial 8:** Diffusion model	

Welcome To Modern AI Summer Workshop!

Why we made this workshop?

- **AI is accelerating:** it's reshaping engineering, biology, and cognition fast.
- **It's genuinely exciting:** On it’s own, modern AI is creative, beautiful, and it can change how you think about problems.
- **For the future AI scientists:** If you want to work *in* AI, you need some depth and coverage, we hope to provide both up to certain degree.
- **For the researchers who'll use it:** if you want to apply AI in your own work, understanding *how* it works matters: knowing what AI can do with your data can change how you design your experiments.

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Our three aims:

- **The big picture:** The main paradigms of AI, and how they connect.
- **Enough depth:** To be able to read and understand recent papers using these approaches.
- **Practical :**get to where you can implement any topic we cover, by yourself, on your own data or provided data.

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Structure of the workshop

- **Modules**: Foundations *(A, B, C, D)* → advanced *(E, F)*

- **Recipe cards**: Step-by-step application of a complex, real ML task, end to end.

- **Tutorials (hands-on):** Implement the core concepts yourself, in practice, on your own data or provided data.

- **Guest lectures:** Applications of AI in cutting-edge research and industry, each linked to the corresponding module.

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What to expect

- **Concepts, visualizations and equations:** We will show the math, but always alongside applications, concepts, visuals, and examples.

- **Something for everyone:** Some of you will might find certain parts easy, others more challenging, depending on your background and prior exposure to math, AI or even biology.
- **We hope everyone takes something out of it**: Whatever your starting point.

- **We will pause for questions at regular intervals,** But we have a lot of materials to cover!

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Sources & inspiration courses

- **Deep Learning:** MIT 6.7960 · Phillip Isola
- **ML with Graphs:** Stanford CS224W · Jure Leskovec
- **Multi-Task & Meta Learning:** Stanford CS330 · Chelsea Finn
- **Deep Generative Models:** Stanford CS236 · Stefano Ermon
- **Flow Matching & Diffusion:** MIT · Peter & Ezra Erives
- **Multimodal ML:** CMU 11-777 · Morency & Liang
- **ML for Bioinformatics:** Sharif University of Technology. Ali Sharifi-Zarchi (in Persian)

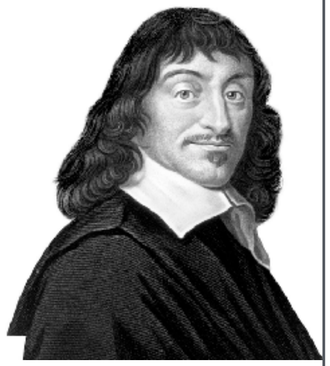
+ books and papers cited throughout the modules

Introduction: A Brief History of AI

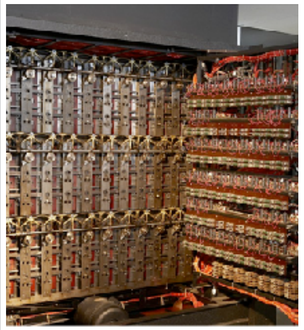
Gauss / Legendre:
learning parameters from data(least squares)

Ancient Myth:
Talos, giant automaton made of bronze

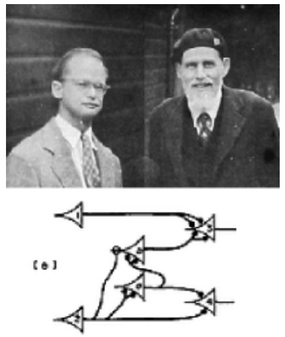
Aristotle:
Formalizing Logic, reasoning, thoughts given by rule



Descartes: *Asked whether a mind is something a machine could ever have*



Boole/Turing:
reduced logic to binary and defined computation



McCulloch/Pitts:
first mathematical model of a neuron

Rosenblatt:
Perceptron, a machine that learn

Hinton/ Linnainmaa/ Werbos/ Rumelhart/ Williams:
Backpropagation,

Hinton/ Krizhevsky/ Sutskever :
Deep learning via Alexnet,

Google Research:
Transformers, LLMs

~700BCE and before

~350BCE

~1637

1805

~ 1900-1940

~ 1943

~ 1958

~ 1986

~ 2012

~ 2017

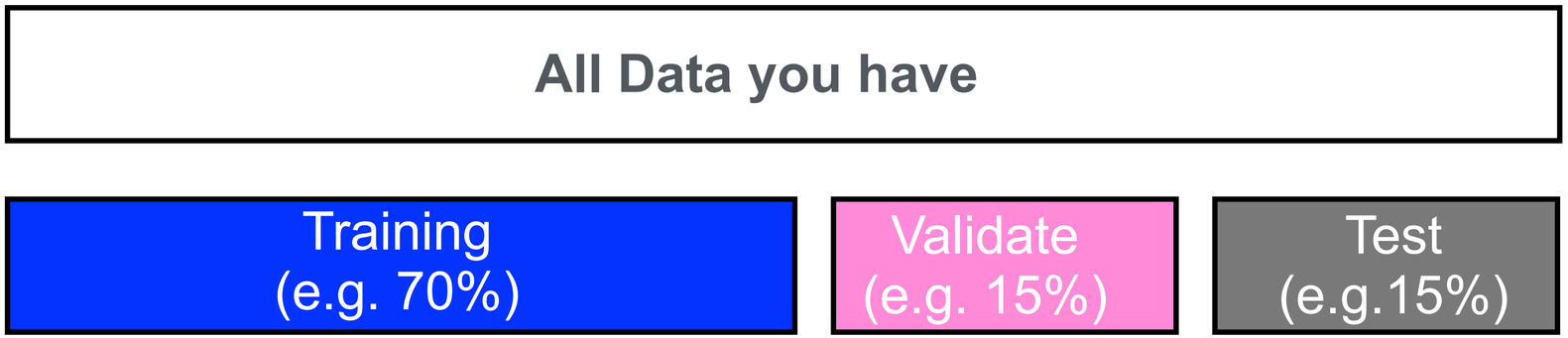
Introduction: A) Basics of Machine learning

Introduction: General workflow of training an AI model

1. Select/develop the model (this course)

AI model

2. Define:
    - The prediction target
    - A training objective (loss)
    - An evaluation metric
3. Divide the data into three parts:
    - Training set: the model learns from this
    - Validation set: tune choices / decide when to stop training
    - Test set: kept hidden while making choices; used only after the final model is fixed.



Introduction: General workflow of training an AI model

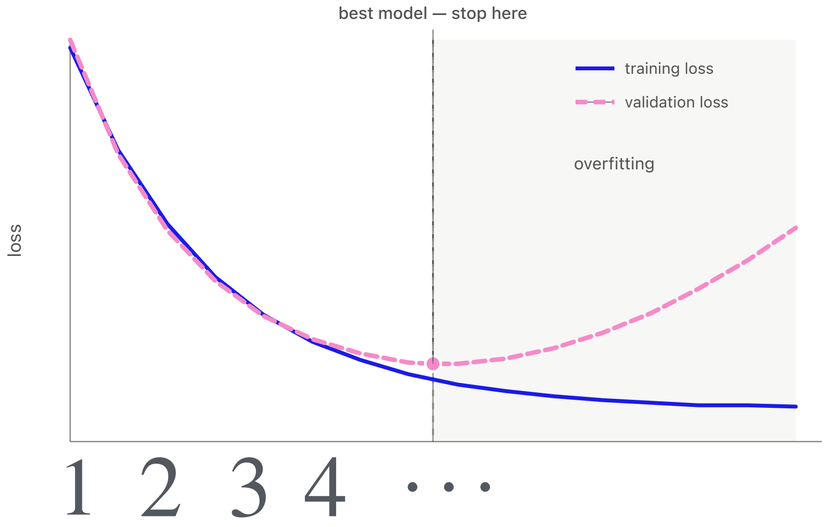
4. Train the model:

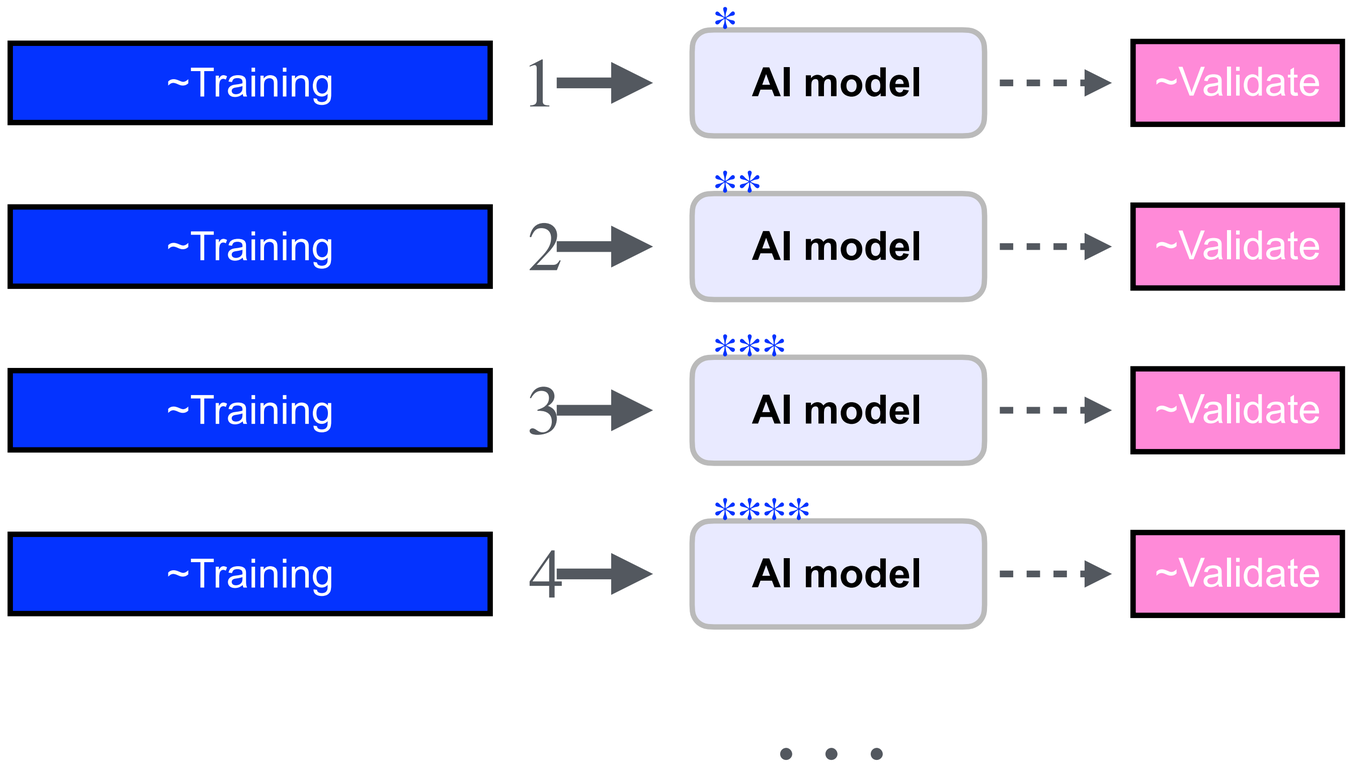
Feed training data through the model.

Use the training loss to update parameters.

Measure validation loss without updating parameters.

Use validation performance to choose settings and decide when to stop.





5. Evaluate via Test data

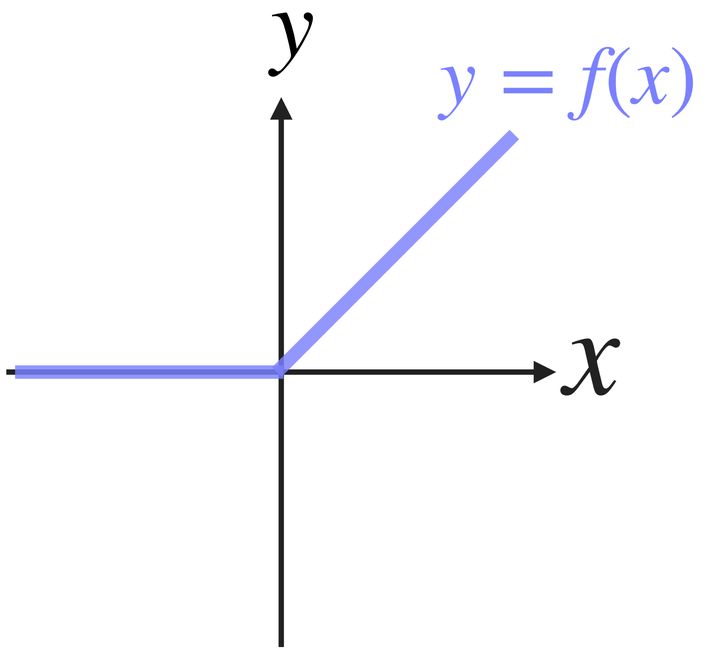
Introduction: B) Gradient descent and backpropagation

Introduction: As simple non linear function

First, consider a simple mathematical function f.

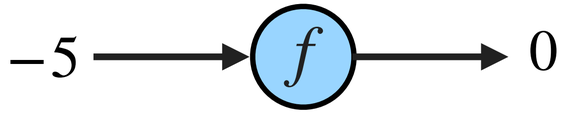
It transforms an input x into an output y:

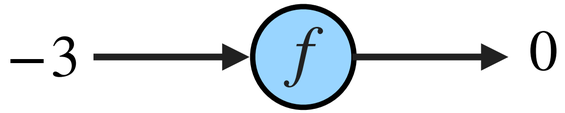
$$y = f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

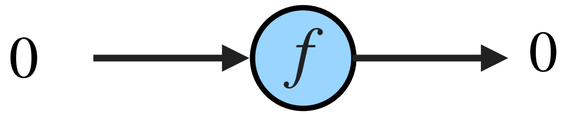


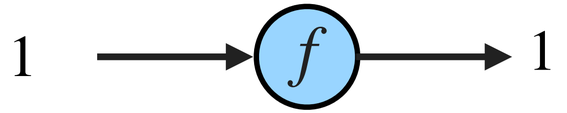
Visually we can show this with the diagram bellow:

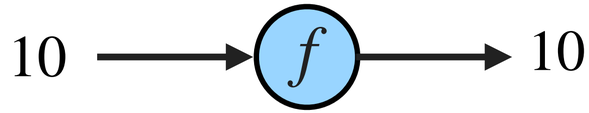
The graph describes how inputs are transformed into outputs. For example:





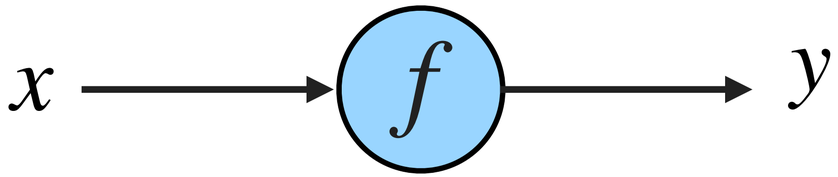






Introduction: Computational graph

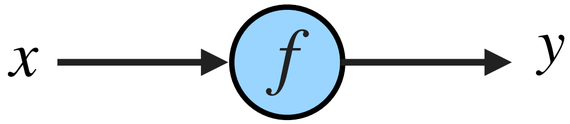
We call this type of drawings, **the computational graph**

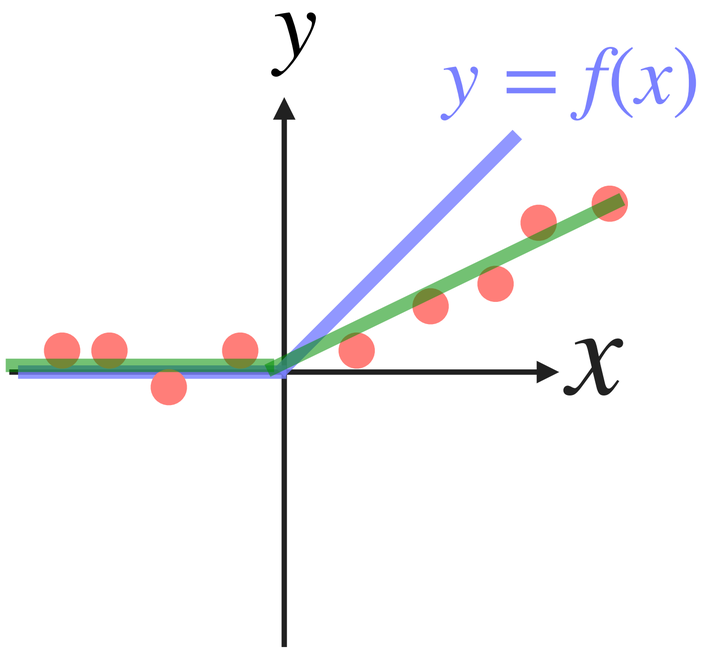


Introduction: As simple non-linear function a predictor

Now suppose we choose the function f to make predictions.

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$





We are given a dataset of input-output pairs:

$Data = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$

Question: Is this function a good predictor for our data?

Not quite.

For $x < 0$, the function predicts values close to the data.

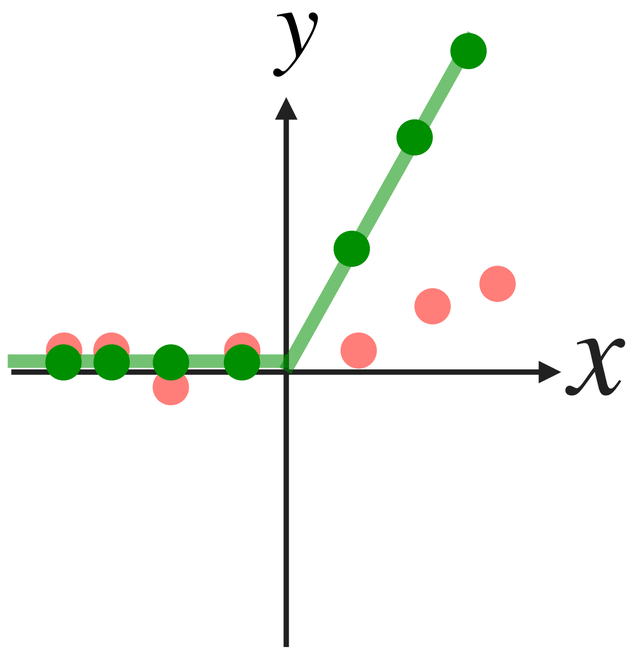
But for $x > 0$, the blue line is too steep.

But this parametrized function is a better predictor when $w = \frac{1}{2}$

$$f_w(x) = \begin{cases} 0 & \text{if } wx < 0 \\ wx & \text{if } wx \geq 0 \end{cases}$$

Introduction: How do we choose the parameter w ?

What if we do not know the best value of w? Can we find it computationally?



We first define a measure of error between the observed data and the model predictions:

$$\text{Data} = \{(x_i, y_i)\}_{i=1}^{n}$$

$$\text{Pred} = \{(x_i, \hat{y}_i)\}_{i=1}^{n}, \quad \text{where } \hat{y}_i = f_w(x_i)$$

We call this measure of error the loss function:

$$L(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Introduction: Choosing w by minimizing the loss

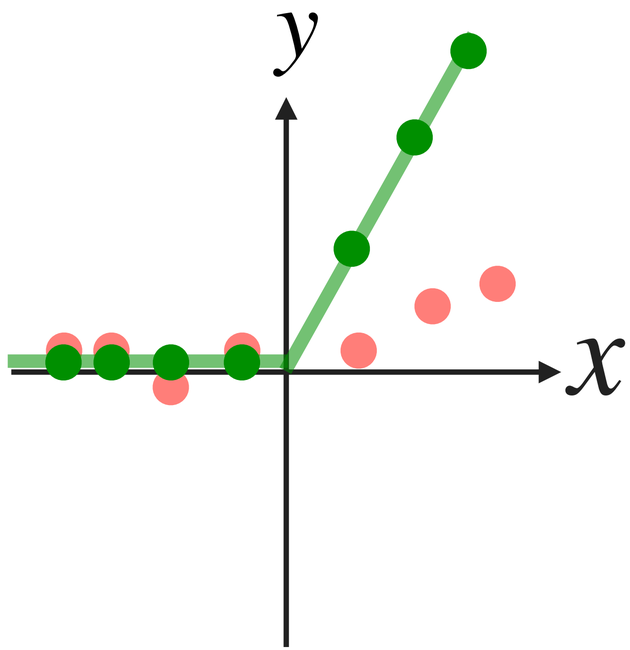
Now that we have a loss function

$$L(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

we can define the best parameter as:

$$w^* = \arg\min_{w} L(w)$$

It means: Find the value of w that makes the prediction error as small as possible.

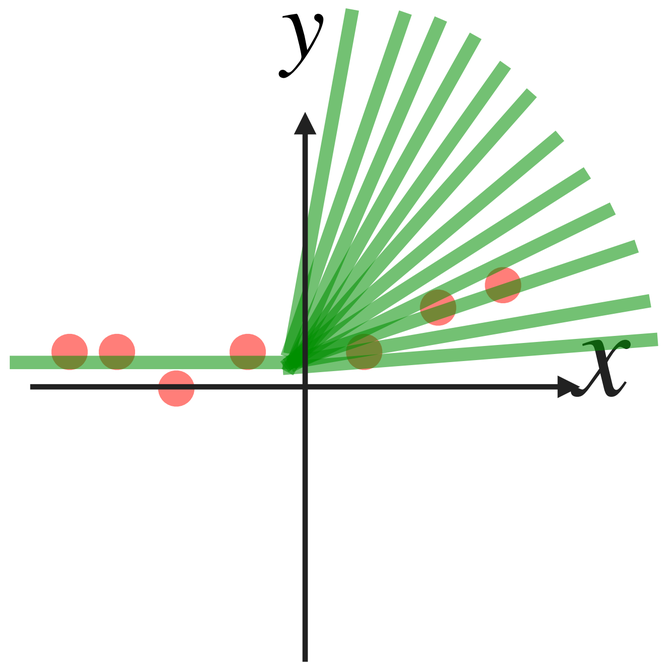


$\text{Data} = \{(x_i, y_i)\}_{i=1}^{n}$

$\text{Pred} = \{(x_i, \hat{y}_i)\}_{i=1}^{n}$,
where $\hat{y}_i = f_w(x_i)$

Introduction: finding w^* by Brute-force search

$$f_w(x) = \begin{cases} 0 & \text{if } wx < 0 \\ wx & \text{if } wx \geq 0 \end{cases}$$



One simple strategy to find w^* is:

Strategy 1: Try many possible values of w, compute the loss for each one, and choose the best:

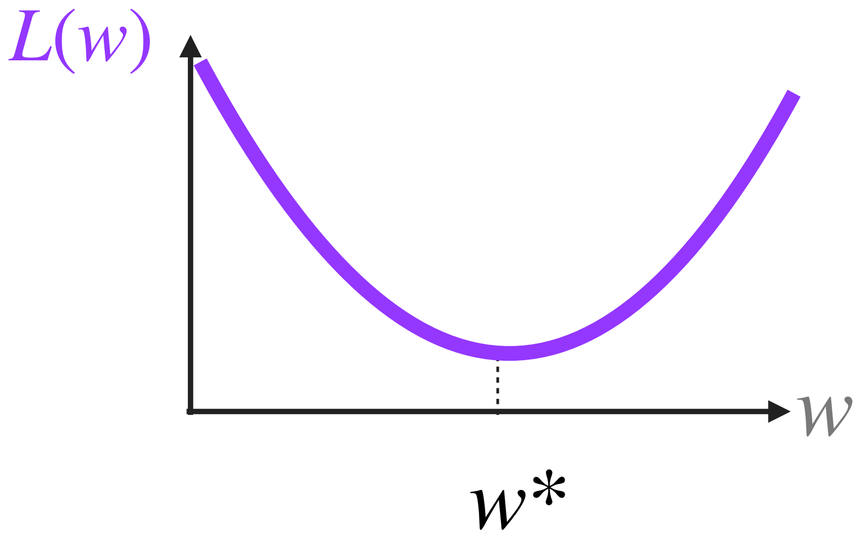
A. Pick a candidate value w.

B. Predict all outputs $\hat{y}_i = f_w(x_i)$

C. Compute the loss $L(w)$

D. Repeat for many values of w.

E. Choose the parameter with the smallest loss, w^* .



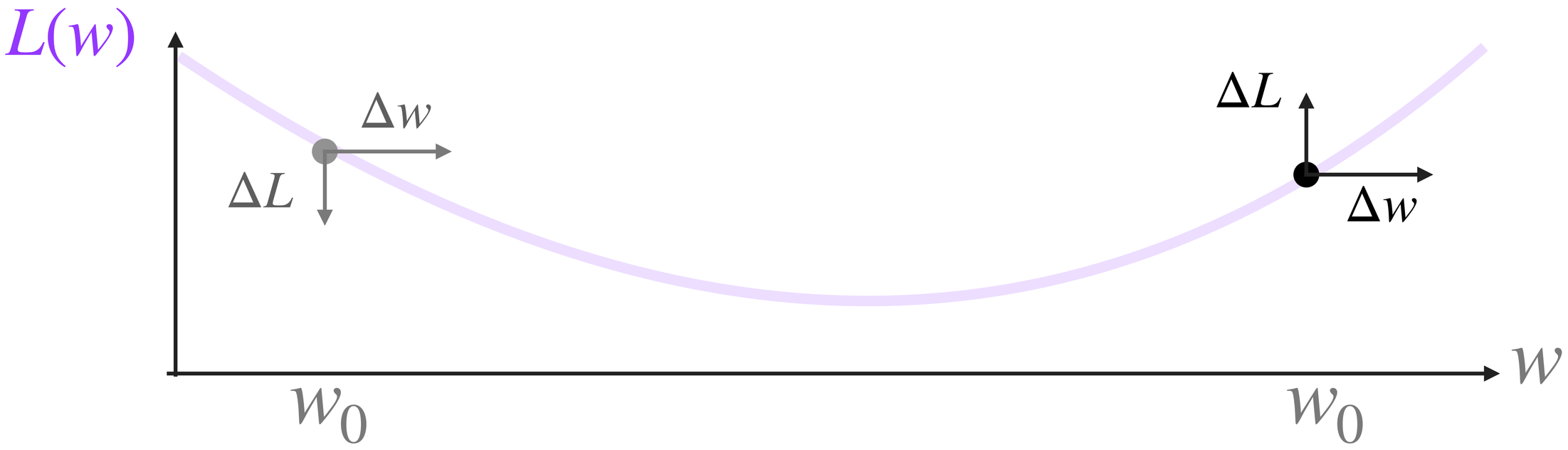
But this strategy is computationally expensive, image when you have 10,000,000 parameter....

Strategy 2: Improving a guess **(gradient descent)**

A. We start from a w_0, calculate $\hat{y} = f_{w_0}(x)$ and $L(w_0)$

B. Try a small change of w towards the positive direction: $w_0 \rightarrow w_0 + \Delta w$

C. Measure what happens to the loss: $\Delta L = L(w_0 + \Delta w) - L(w_0)$



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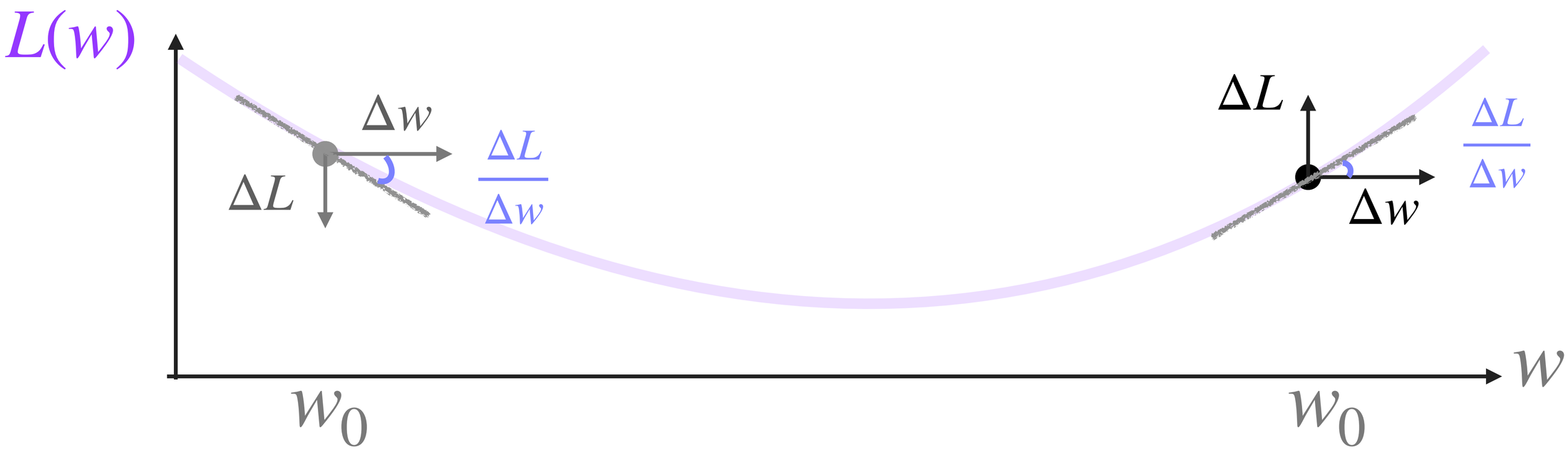

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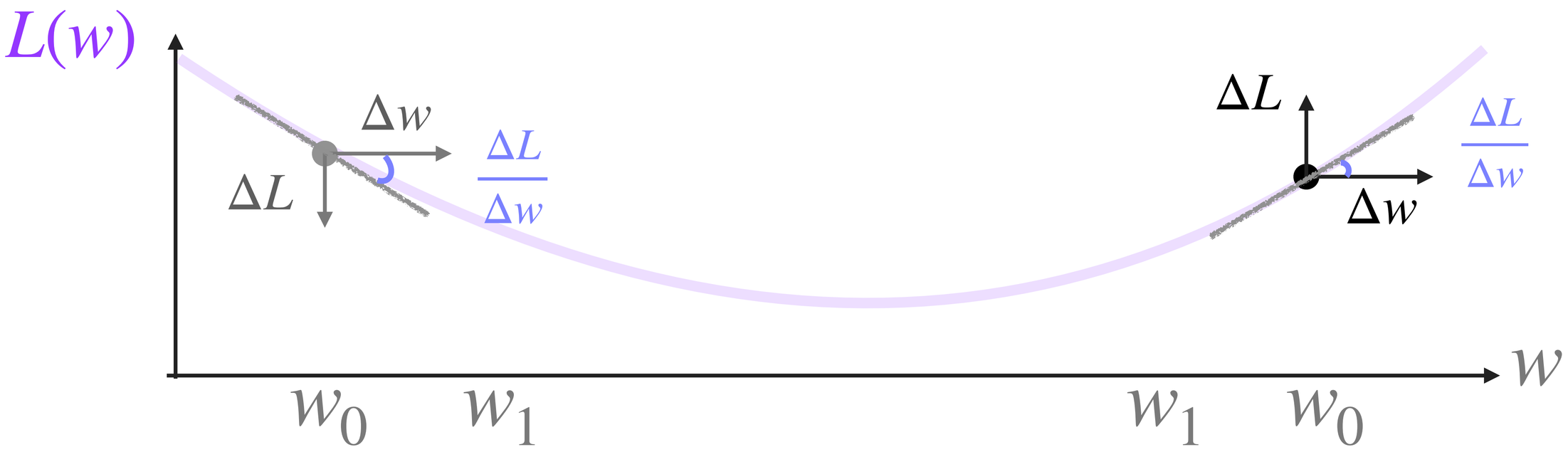
C. Measure what happens to the loss: $\Delta L = L(w_0 + \Delta w) - L(w_0)$

D. Estimate the local slope of the loss: $\frac{\Delta L}{\Delta w}$

E. If the slope is positive, move left. If the slope is negative, move right: $w_1 = w_0 - \eta \frac{dL}{dw}$ (η is the learning rate)



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A. We start from a w_0, calculate $\hat{y} = f_{w_0}(x)$ and $L(w_0)$

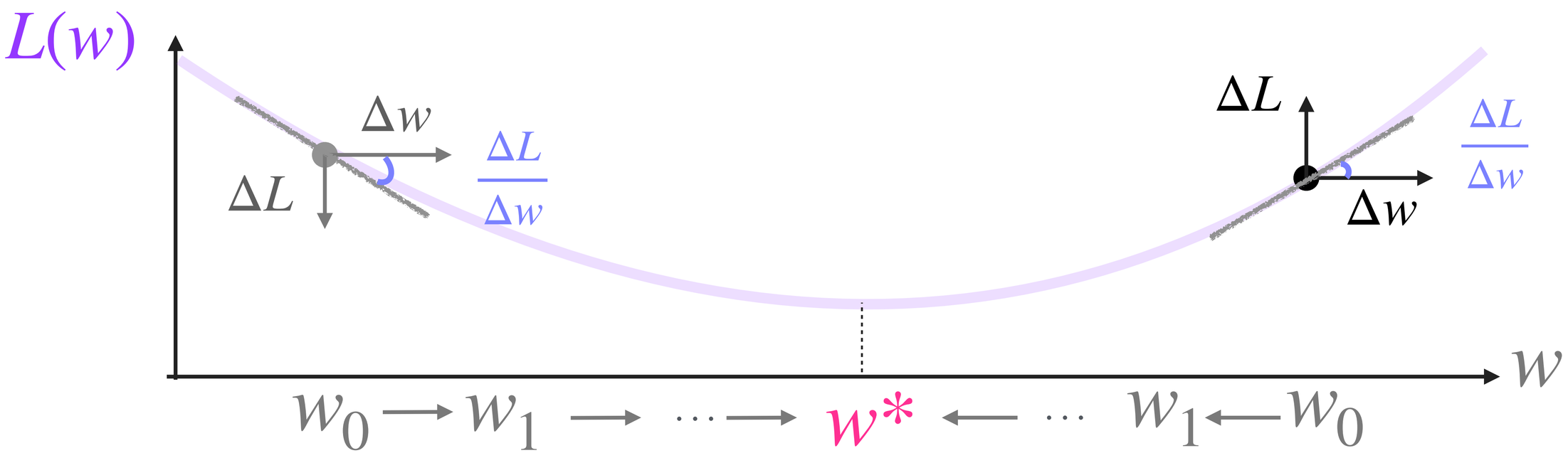
B. Try a small change of w towards the positive direction: $w_0 \rightarrow w_0 + \Delta w$

C. Measure what happens to the loss: $\Delta L = L(w_0 + \Delta w) - L(w_0)$

D. Estimate the local slope of the loss: $\frac{\Delta L}{\Delta w} \approx \frac{dL}{dw}$

E. If the slope is positive, move left. If the slope is negative, move right: $w_1 = w_0 - \eta \frac{dL}{dw}$ (η is the learning rate)

F. Repeat until the loss stops decreasing

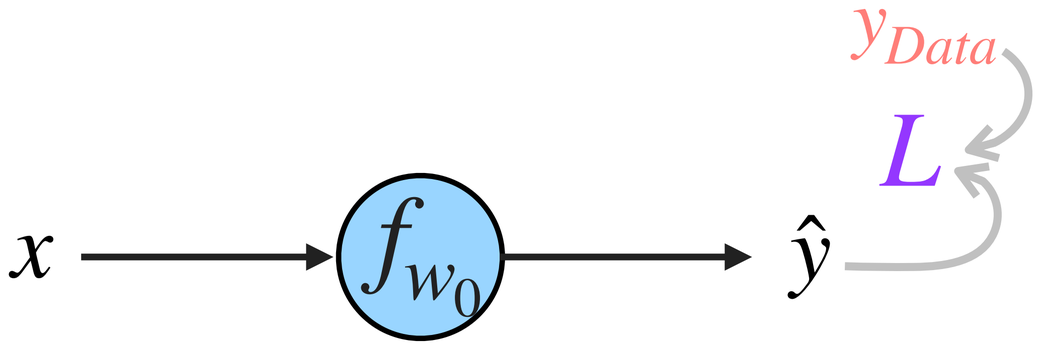


Computational graph of gradient descent algorithm

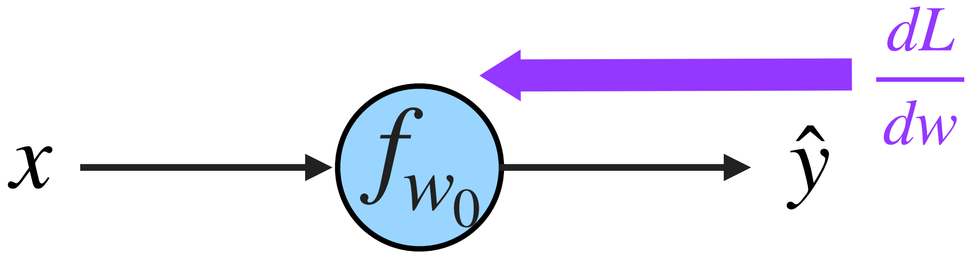
Strategy 2: Gradient descent algorithm:

Step1, Forward: Start from w_0 , calculate

$$\hat{y} = f_{w_0}(x) \quad \text{and from that,} \quad L(w_0)$$



Step2, Backward: Calculate $\frac{dL}{dw}$ at $w = w_0$



Step3, Update: The new updated weight is

$$w_1 = w_0 - \eta \frac{dL}{dw} \quad (\eta \text{ is the learning rate})$$

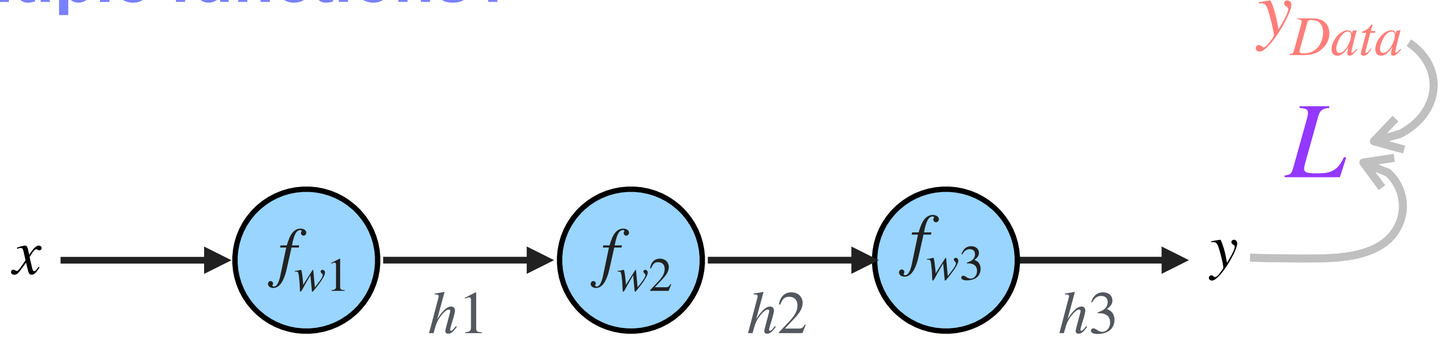
Step4: Repeat until the loss stops decreasing

Bonus:

Introduction: What if we have multiple functions?

Step1, Forward:

$$h_1 = f_{w1}(x)$$
$$h_2 = f_{w2}(h_1)$$
$$h_3 = f_{w3}(h_2) = \hat{y}$$
$$L = L(y, \hat{y})$$



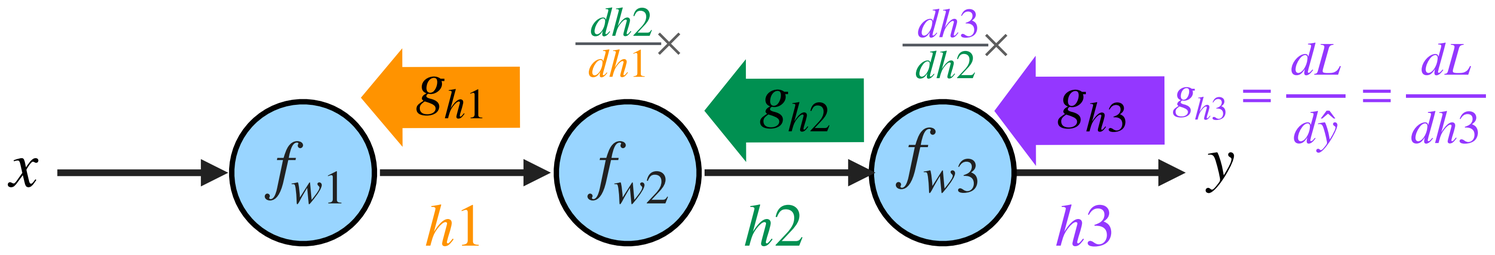
Step2, Backward: Let g be the backward gradient signal.

At the output $g_{h3} \equiv \frac{dL}{dh3}$

$$\frac{dL}{dw3} = g_{h3}\frac{dh3}{dw3} \quad \textbf{and} \quad g_{h2} = g_{h3}\frac{dh3}{dh2}$$

$$\frac{dL}{dw2} = g_{h2}\frac{dh2}{dw2} \quad \textbf{and} \quad g_{h1} = g_{h2}\frac{dh2}{dh1}$$

$$\frac{dL}{dw1} = g_{h1}\frac{dh1}{dw1}$$

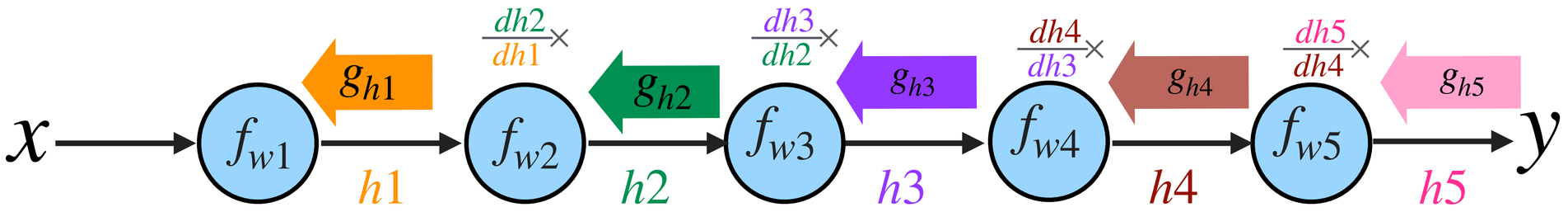


Step3, and **Step 4** are the same as before

Bonus:

Introduction: What if we have multiple functions?

This pattern can go as deep as you want, and we just need to compute each local gradient g only once

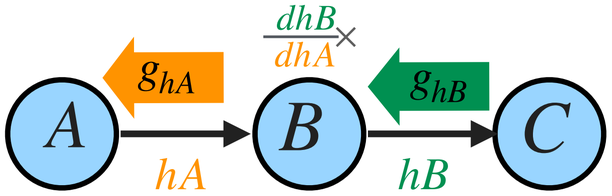


$$g_{h5} = \frac{dL}{d\hat{y}} = \frac{dL}{dh5}$$

Bonus:

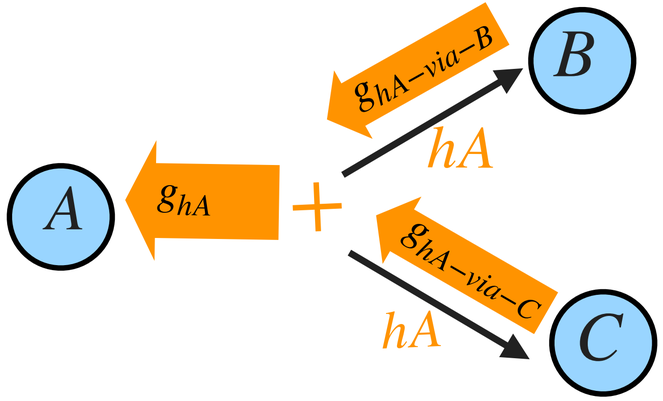
Introduction: What if we have multiple functions?

Sequential computation (chain rule)



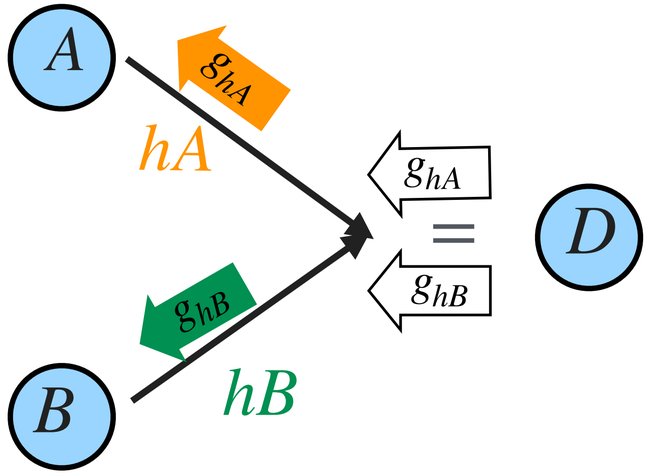
Gradient:
multiply the gradient by the local derivative.

One value feeds multiple functions (split)



Gradient:
gradients from all outgoing branches are added at the split point.

Multiple values feed one function (join)



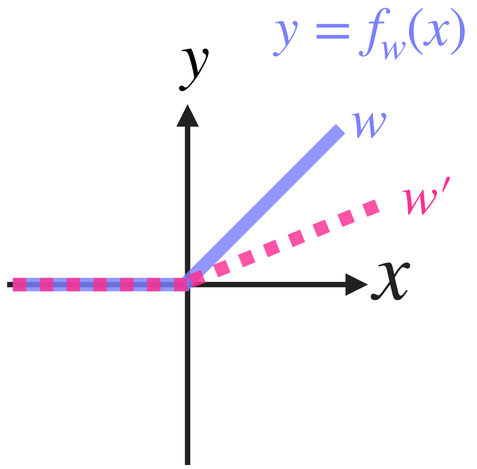
Gradient:
Each input receives a copy gradient from the source.

Bonus:

Introduction: Adding a second learnable parameter to a simple non linear function

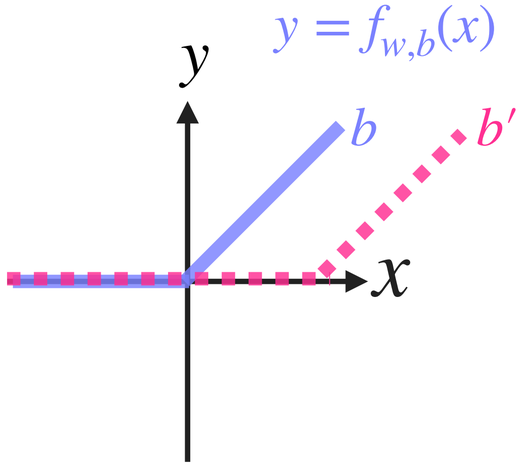
Previously, our function had one parameter, w:

$$f_w(x) = \begin{cases} 0 & \text{if } wx < 0 \\ wx & \text{if } wx \geq 0 \end{cases}$$



Now we introduce a second learnable parameter, b:

$$f_{w,b}(x) = \begin{cases} 0 & \text{if } wx + b < 0 \\ wx + b & \text{if } wx + b \geq 0 \end{cases}$$



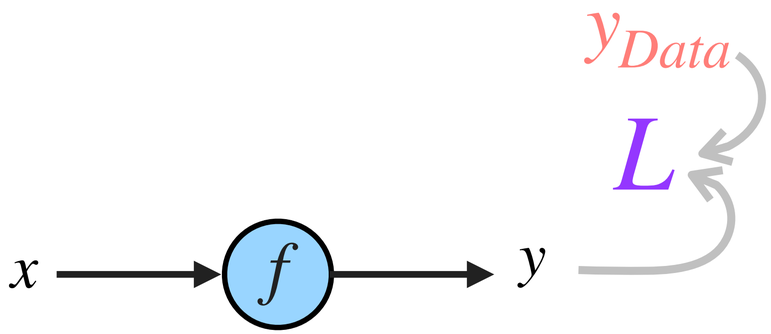
This make our function more flexible

Bonus:
Closing: What we learned

We saw four pieces:

1. A model is a computational graph.

$$y = f(x)$$



2. A loss measures prediction error.

$$L(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

3. Gradient descent updates function parameters.

$$w = w - \eta\frac{dL}{dw}$$

Iteratively update w:

forward pass → loss → backward pass → update → repeat

4. Backpropagation computes gradients efficiently, when we have connected functions